

Seminar topic: **Software Synthesis for Hardware Processors**

Abstract

Optimized low-level software code, synthesized from high-level behavioral specifications, ensures efficient and accurate program execution on hardware by offsetting resource limitations. As hardware and software are parallelly developed to fulfill system-level goals, software synthesis can pave the path for early planning of the significant challenge in embedded field- the hardware constraint; thus, can improve upon performance, cost-efficiency and flexibility across both domains. C/C++/SystemC are the most common languages used for code automation. The targeted back-end hardware like MCUs, DSPs, FPGAs and AISPs are significant beneficiaries of the synthesis procedures. Wherever design parameters like power consumption, real-time deadlines, memory allocation are of critical concerns, hardware software seamless alignment is highly anticipated.

The goal of the seminar paper is to explore some of the core aspects of software synthesis like scheduling tasks to processors. It will provide a comprehensive understanding of efficient mapping of software onto embedded processors while adhering to system-level critical constraints including timing, resource limitations and power consumption.